



School of Public Health

Biostatistics for Bachelor of Pharmacy students

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2. Methods of data collection

2.1 Sources of data, methods of data collection and presentation

2.2. Data types and measurement scales

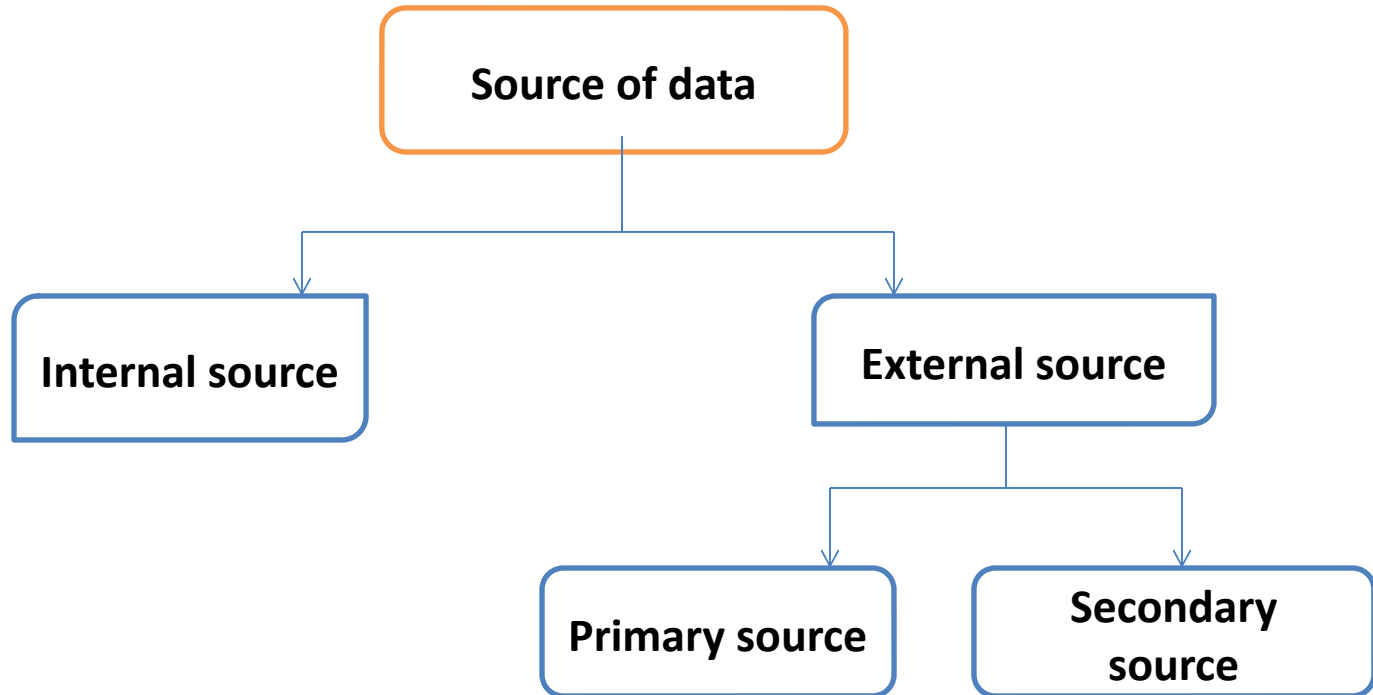
2.3. Data collection methods

2.4 . Questionnaire design and interviewing techniques

Objectives

- **At the end of this session, students are expected to:**
- List sources of data
- Describe the advantage and disadvantages of each data collection methods
- Apply appropriate method of data collection in real clinical scenario
- Explain the characteristics of Good questionnaire
- Develop well structured questionnaire

Source of Data



Internal and External Source of Data

Internal Sources of Data

- Many institutions and departments have information about *their regular functions*, for their own internal purpose.
- When those information is used in any *survey*, it's called Internal Source Data.
- E.g., Public health Institutes & Nursing association members etc.

External Sources of Data

- When information is collected from *outside agencies*, it is called external source of data.
- Such type of data are either *Primary or Secondary*.
- This type of information can be collected by *Census* or Sampling method by conducting *surveys*

Primary Data

- Primary data are those which are collected for the **first time**.
- It is real time data which are collected by the **researcher** himself.
- This is the process of collecting and making use of the data.
- This data originated by the researcher specifically to address the research problem

Secondary Data

- Secondary data are those that **have already** been collected by others.
- These are usually in journals, periodicals, research publications, official records etc.
- Secondary data may be available in the published or unpublished form.
- When it is not possible to collect the data by primary method, the investigator go for secondary method.
- This data collected for some purpose other than the problem at hand.

Difference between primary and secondary Data

Primary Data

- Real time data.
- Sure about sources of data.
- Help to give results/finding
- Costly and Time consuming process.
- Avoid biasness of response data
- ~~More flexible.~~

Secondary Data

- Past data.
- Not sure about sources of data.
- Refining the problem.
- Cheap and No time consuming process.
- Can not know in data biasness or not
- Less Flexible.

Systems of data collection

There are two systems of data collection

1. Regular system- registration of events as they become available
2. AD-hoc system- it is a form of survey to collect information that is not available

Methods/techniques of data collection

- Before any statistical work, data must be collected.
- Data collection techniques allow us to systematically collect data about our objects of study (people, objects, and phenomena) and about the setting in which they occur.
- In the collection of data we have to be systematic.
- If data are collected haphazardly, it will be difficult to answer our research questions in a conclusive way, and
- Before the actual data collection the **reliability and validity** of the tool should be checked

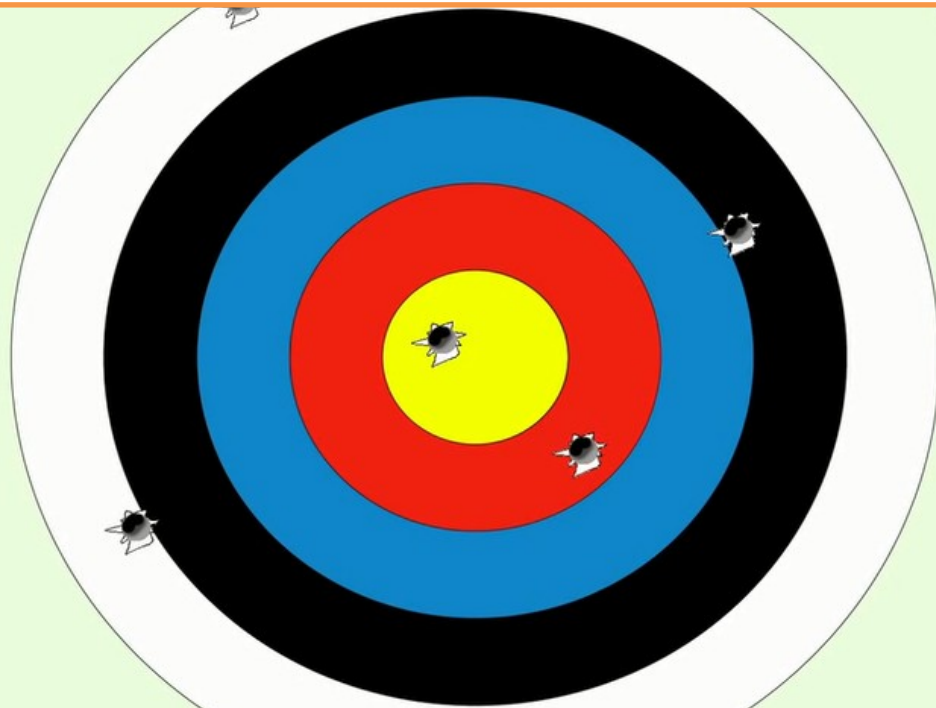
Reliability and Validity of measurement

- **Reliability** refers to the consistency or dependability of a measurement technique
- More specifically, reliability is concerned with the consistency or stability of the score obtained from a measure or assessment technique over time

Reliability and Validity...

- **Validity**: The concept of *validity* refers to *what* the test or measurement strategy measures and *how well* it does so.
- Conceptually, validity seeks to answer the following question:
- “***Does the instrument or measurement approach measure what it is supposed to measure?***”

Unreliable



Unreliable

Valid and reliable



Reliable but not valid



Data collection methods...

- Data can be collected in a variety of ways
- Factors influencing choice of data collection methods are
 - Type of data/variable
 - Objective of the study
 - Availability of data
 - Money
 - Time
 - Quality needed
 - Characteristics of study participants

Data collection methods...

Methods of data collection

- Formal testing (psychological, educational, academic, intelligence)
- Observation
- Face-to-face and self-administered interviews
- Postal or mail method and telephone interviews
- Using available information
- Focus group discussions (FGD)
- Other data collection techniques – Rapid appraisal techniques, life histories, case studies, etc.

Observation

- Systematically selecting, watching and recoding behaviors of people or phenomena
- Give additional, more accurate information on behavior than interviews or questionnaires
- Can be used in any kind of research (quantitative, qualitative)
- It can be simple visual observations or
- high level using sophisticated machines and equipment such as biochemical, X-ray machines, microscope, clinical and microbiological examinations.
- Efficient way when researchers studies & quantifies some type of behavior.

Observation ...

- There should be standardized outline/guidelines for the observations prior to actual data collection to ensure adequate validity and inter-rater reliability
- The oldest and most basic tool
- Data': body language, facial expression, behavior, other non-verbal expressions, movements, etc.
- Observation of behaviors, actions, activities and interactions helps in understanding more than what people say about situations, and complex situations more fully.

Advantage

- Provides deep understanding and relatively more accurate data on behavior and activities
- Permit researcher/evaluator to enter into and understand situation/context.
- Provide direct information about behavior of individuals and groups.
- Allows to observe whether people do what they say they do; often used for exploring deviant behavior
- Useful to access tacit knowledge of subjects - the subconscious knowledge that they would not be able to verbalize in an interview setting

Disadvantages

- Time consuming and expensive
- Observer bias: selective perception of observer may distort/affect the investigation.
- Ethical issues concerning confidentiality or privacy may arise.
- May affect behavior of participants
- Need well qualified, highly trained observers; may need to be content experts during the use of high level machines and good skills in research and local language;
- Time lag between observation and note taking is likely thus requires good memory and ability to take note afterwards

Interviews and Self administered Q.

- ✓ The most commonly used data collection techniques.
- ✓ Once the decision has been made to use these techniques, the following questions should be considered before designing our tools:
 - ✓ What exactly do we want to know, according to the objectives and variables we identified earlier?
 - ✓ Of whom will we ask questions and what techniques will we use?
 - ✓ Are our informants mainly literate or illiterate?
 - ✓ How large is the sample that will be interviewed?

Self-administered questionnaire

- Written questions are presented that are to be answered by the respondents in written form
- A written questionnaire can be administered in different ways, such as
 - Sending questionnaires by mail
 - Gathering all or part of the respondents in one place at one time, giving oral or written instructions, and letting them fill out the questionnaires
 - Hand-delivering questionnaires to respondents and collecting them later

Self administered Q....

- ✓ **Self-administered** questionnaires is **simpler** and **cheaper**;
- ✓ They demand a certain level of education and skill of the respondents;

Interview

- Oral questioning of respondents to tell us their story in their own ways assuming their perspectives are meaningful.
- Relatively simple approach, but it can produce a wealth of information
- Relatively inexpensive and efficient way to collect a wide variety of data that does not require formal testing.
- The effectiveness of an interview depends on how it is structured
- Interviewer should be trained properly to avoid variation of collected data

Interview ...

- Interviews can be conducted with varying degree of flexibility

High degree of flexibility: when

- Studying sensitive issues (e.g. commercial sex)
- The researcher has little understanding of the problem
- Is frequently applied in exploratory studies

Low degree of flexibility: when

- Researcher is relatively knowledgeable about expected answers or
- The number of respondents being interviewed is relatively large

- Questionnaires may be used with a fixed list of questions in a

Types of interview

In-depth Interviews

- Dialogue between a skilled interviewer and an interviewee to elicit rich, detailed material that can be used in analysis
- Best conducted face to face,
- The interviewer does not follow a rigid form rather limited set of open-ended questions with extensive probing
- Seek to encourage free and open responses in their own words
- Interview Guide need to be prepared.

Conducting an In-depth Interview

Interview Guide

- ✓ Includes a list of questions or issues that are to be explored
- ✓ Helps the interviewer to remember the points to cover
- ✓ Suggests ways of approaching and talking about topics make interviewing more systematic and comprehensive.
- ✓ Reminds the interviewer about probes and ways of asking questions

- ✓ Gives a possible order of topics; and
- ✓ Helps the interviewer to enable people to talk in their own ways, and fully as possible.
- ✓ Includes an introduction and way of ending the interview.

Conducting an In-depth ...

When to use In-depth interviews

- Highly sensitive issues
- To get detailed information about a person's thoughts and behaviors
- When peer pressure is expected to distort facts

Cont'd...

Important points to be remember

- Avoid variations in the interview setting
- informed consent of the respondent should be obtained
- Knowledge gap between the interviewer and interviewee must be considered to facilitate understanding
- Use a tape record and or a note taker who can assist
- Confidentiality of information must be maintained
- Receive the information accurately

➤ Information can be distorted by interviewer

- Fatigue
- Bias (expectation of answers),
- Preoccupation with taking notes and
- By technical languages (Jargons) foreign to the interviewee

Tips for Effective Interviewing

- Follow up on what the participant says
- Ask questions when you don't understand
- Ask to hear more about a subject
- Listen more, talk less, and ask real questions
- Ask open-ended questions
- Follow up, don't interrupt
- Ask participants to tell a story
- Keep participants focused, and ask for details
- Avoid reinforcing your participants' responses
- Use an interview guide carefully
- Tolerate silence

Advantages

- Usually yield richest data, details, new insights;
- Permit face-to-face contact with the respondents;
- Provide opportunity to explore topics in depth;
- Permits clarification of questions.
- Affordability to experience the affective as well as cognitive aspects of responses
- Allow interviewer to explain or help clarify questions, increasing the likelihood of useful responses.
- Suitable for use with both literates & illiterates.
- Allow the interviewer to be flexible in administering interview to particular individuals or circumstances

Disadvantages

- Expensive and time consuming
- Data obtained may not be appropriate for extensive statistical analysis because they simply describe a construct rather than quantifying it
- Need well qualified, highly trained interviewers;
- Interviewee may distort information through recall error, selective perceptions,
- Presence of interviewer can influence, desire to please interviewer
- Flexibility can result in inconsistencies across interviews
- Volume of information too large, may be difficult to transcribe & reduce data

Telephone interview

- They are less costly
- People may be more candid in their opinions, since there is no face to-face contact.
- A major drawback to the telephone survey is that some people may not have phones or may not answer when the calls are made. Chance of being surveyed??
- Unlisted numbers and cell phones, so they cannot be surveyed.
- Tone of the voice of the interviewer might influence the response of the person who is being interviewed.
- We may not get a rich and detailed responses

Mail interview

- Cover a wider geographic area than telephone in-depth interviews
- Less expensive to conduct.

Disadvantage

- Low number of responses and inappropriate answers to questions.
- People may have difficulty reading or understanding the questions.

Use of documentary sources/available information

Available information includes

- ✓ Clinical and other personal records,
- ✓ Published mortality statistics,
- ✓ Census publications,
- ✓ Publication of Ministry of Health and Other Ministries
- ✓ News Papers and Journals.
- ✓ International Publications like Publications by WHO, World Bank, UNICEF
- ✓ Records of hospitals or any Health Institutions.
- ✓ Use of key informants is important technique to gain access to available information

Advantages

- Ease of access
- Low cost or free/ collection is inexpensive
- Time-saving
- Allow you to generate new insights from previous analysis
- Longitudinal analysis
- A huge amount of secondary data with a wide variety of sources

Disadvantages

- Might be not specific to your needs
- You have no control over data quality
- Data may not always be complete and precise enough, or too disorganized
- Biasness
- Not timely – collected before and can be out - dated
- You are not the owner of the information

Focus group discussion (FGD)

- FGD is a qualitative data collection method of approximately 6-10 people guided by a facilitator and discuss an issue/research topic among themselves in depth
- People with something in common discussing an issue /Homogenous group composition/
- It is focused Ideal size 6-10, maximum =12, minimum=4
- Far advanced and used more than in-depth interview, because
 - Group interaction stimulate richer responses and emergence of new ideas
 - Observation: The researcher observes and gets first hand insights
 - Cost and timing: (FGD) can be done more quickly and generally less expensively than series of depth interviews

FGD...

It requires

- Relaxed, informal/natural setting
- Time duration: 1:30 - 2:00 hours (it depends)
- One leader (moderator) , one assistant (or co-moderator)
 - Good communication skills
 - Records all the verbal and non-verbal message, in addition to note
- FGD guide: tool used to guide the discussion, probe/follow up questions

FGD...

When to use FGD

- Idea generation
- Problem identification and definition
- Evaluation of message concepts
- Program design

FGD... Advantages

- Produce a lot of information in less time and cost /Richness of data/
- Excellent for obtaining information from illiterate communities
- Explore variety of opinions or views within a group
- Flexibility in covering topics/May uncover unanticipated ideas that are important
- Broader understanding by providing well-grounded data on social and cultural norms with respect to health and disease, prevention of disease
- Ability to study special respondents
 - youth
 - Professionals (doctors, lawyers)

Disadvantages

- Needs more time for preparations
- Requires good facilitation skills
- Lack of generalizability (small sample size)
- High selection bias
- Might be misused
- FGD is not a replacement for quantitative research
- Subject to Interpretation - subjective
- Cost-per-respondent is high (compared to survey)
- Results dependent on skill of moderator in running the group and analysis (researcher)

Good facilitation skills

- Build rapport and open the discussion with a general idea
- Give participants a chance to think about the questions/do not lookdown on respondents
- Covering the material in the guide
- Do not correct participants during discussions/a moderator is not a teacher/judge/
- Turn the group away from irrelevant areas.
- Shows flexibility; e.g order of guide
- Links ideas together
- Encourages participation from everyone

Good facilitation skills...

- Treating the participant as the expert?/ does not agree or disagree with what is said, no right or wrong answers
- Active listening; Pay attention to what participants say and follow up with relevant questions and probes
- Don't lose control
- Build rapport with and among participants; it starts from welcoming
- Prevent dominance

Skills Required for Moderator

- Observation
- Interpersonal
- Communication
- Interpretive

Reading assignment

- Rapid appraisal technique , Delphi technique , nominal group discussion, life history and case study of data collections??

Questionnaire

- **A questionnaire (or form)** is a group or sequence of questions designed to obtain information from a respondent.
- Questionnaires include forms used for sample surveys and censuses as well as administrative forms.
- Questionnaires play a central role in the data collection process.
- They have a major impact on data quality since the questionnaire is the process whereby data are collected

Importance of questionnaire

- Is to provide a standardized interview across all subjects.
- They are always asked in exactly the same way.
- **A well-designed questionnaire should:**
 - 📊 Collect data efficiently with a minimum number of errors and inconsistencies;
 - 📊 Be respondent friendly and interviewer friendly (if interviewer-assisted);
 - 📊 Lead to an overall reduction in the cost and time associated with data collection

Types of questionnaires

- ❑ ***Depends on how the questions are asked and recoded, there are two types of questionnaire***

1. Open-ended Questions

- The respondent is not given any possible answers to choose from.
- Permit free responses that can be recorded in the respondent's own words.
- It is used in Facts with which the researcher is not very familiar; Opinions, attitudes, and suggestions of informants, or Sensitive issues

Types of questionnaires...

- Example: Can you describe exactly why girls join sex work?
- What do you think about the public attitude towards COVID-19?
- What do you think are the reasons for low Entrance exam achievement of Grade 12 students ?

Types of questionnaires...

2. Closed Questions

- Offer a list of possible options or answers from which the respondents must choose.
- choices should be exhaustive and mutually exclusive
- Keep the number of options as few as possible.

Types of questionnaires...

- Closed questions are useful if the range of possible responses is known.
- For example: “What is your educational status?”
- 1. can't read and write
- 2. From grade 1-8
- 3. From grade 9 to 12
- 4. Diploma and Above

Types of questionnaires...

- Closed questions may also be used if one is only interested in certain aspects of an issue and does not want to waste the time of the respondent and interviewer by obtaining more information than one needs.
- Did you eat any of the following foods yesterday?

(Circle yes or no for each set of items)

✓ Peas, bean, lentils	Yes	No
✓ Fish or meat	Yes	No
✓ Eggs	Yes	No
✓ 09/25/2026 Milk or Cheese	Yes	No

Questionnaire Design

General Considerations

- With only a few exceptions, long questionnaires get less response than short questionnaires.
- Ask only questions that directly address study goals
- Involve experts in the questionnaire design process
- Formulate a plan for statistical analysis during the design stage of the project

Questionnaire design...

Points to be taken into account when we design a questionnaire

- Keep questions short and simple (KSS)
- should be unambiguous & not double barreled
- Use simple and direct language.
- Wording of a question should be simple & to the point
- To reduce misunderstanding, make items brief
- The best kinds of questions are those allow a pre-printed answer to be ticked

Questionnaire design...

- Leading questions **shouldn't** be asked
- The questions should fall into a logical sequence
- It should be translated into local languages to be used
- The last step in questionnaire design is to test the questionnaire with a small number of interviews before conducting your main interviews by **pre-test/pilot**

Common problems in designing questionnaire

- ✓ Language barriers
- ✓ Lack of adequate time
- ✓ Expense
- ✓ Inadequately trained and experienced staff
- ✓ Invasion of privacy, Suspicion (mistrust)
- ✓ Bias: (Systematic error (not random) in a study that leads to an incorrect estimate of the association between exposure and disease)
- ✓ Cultural norms (e.g. which may preclude (prevent) men interviewing women)

Steps in designing questionnaire

CONTENT: Take your **objectives** and **variables** as your starting point.

FORMULATING QUESTIONS: Formulate one or more questions that will provide the information needed for each variable. Here consider the following points

- ✓ Check whether each question measures one thing at a time.
- ✓ Example, 'what type of question and your classmates prefer?'
- ✓ It is better be divided into two questions because you and your classmates may have different opinions on the preferred interval.

Steps in designing questionnaire...

- ✓ Avoid leading questions: A leading question suggests a certain answer.
- ✓ Example: "Do you **agree** that the district health team should visit each health center **monthly**?" this hardly leaves room for “no” or for other options.
- ✓ Better: “Do you think that district health teams should visit each health center? If yes, how often?”

SEQUENCING OF QUESTIONS: Design your interview questionnaire to be “**consumer friendly.**”

- ✓ Sequence the questions in logical order; background, context
- ✓ Use simple language
- ✓ Put sensitive questions at the end

Steps in designing questionnaire...

Formatting the questionnaire.

- ✓ Your questionnaire should not only be consumer but also **user friendly!**

Translation: If interview will be conducted in one or more local languages, the questionnaire has to be translated to **standardize** the way questions will be asked.

- ✓ After having it translated you should have it **retranslated** into the original language. You can then compare the two versions for differences and make a decision concerning the final phrasing of difficult concepts.

- **Data Organization and Presentation**

Objectives

- **At the end of this section students are expected to**
 - List the methods of data presentation with their advantage and limitation
 - Choose the appropriate methods of data organization for a given problem
 - Apply appropriate methods of data presentation in health and health related conditions

Methods of data organization and presentation

- The data collected in a survey is called **raw data**.
- In most instance useful information is **not immediately evident from the mass of unsorted data**.
- Collected data need to be organized in such a way as to condense the information they contain in a way that will show patterns of variation clearly

Organization and Presentation of data

- Having collected and edited the data, the next important step is to organize it.
- The process of arranging data in to classes or categories according to similarities is called **classification**
- Classification is a **preliminary and it prepares the ground** for proper presentation of data.
- The pictorial presentation of data is broadly classified in to the following two categories:
 - **Tabular presentation**
 - **Diagrammatic and Graphic presentation.**

1. Tabular presentation of Data

Importance of statistical Tabulation

- Statistical data arranged in tables have some definite advantages over those descriptively stated.
 - Tabulated data can be **easily understood** than facts stated in the form of description
 - They have a **lasting impression**
 - They **facilitate comparison**
 - Statistical tables make easier the **summation** of items and detection of errors and omissions
 - When data are tabulated all **unnecessary details** and **repetitions** are **avoided**.

Parts of a table

- a) Title
 - b) Captions
 - c) Stubs
 - d) Body
 - e) Head note
 - f) Foot note
 - g) Source
- Most of the times, the 1st four parts are present in all tables while the presence of the remaining three depends upon the specific purpose.

Parts of a table cont...

- a) **Titles** : It explains
- What the data are about
 - from where the data are collected
 - time period of the data
 - how the data are classified
- b) **Captions**: The titles of the columns are given in captions.

In case there is a **sub-division of any column** there would be **sub-caption** headings also.

- c) **Stubs**: The titles of the rows are called stubs
- d) **Body**: Contains the numerical data

Parts of a table cont...

- e) **Head note:** A statement below the title which clarifies the content of the table
- f) **Foot note:** A statement below the table which clarifies some specific items given in the table **e.g.** it explains omissions, etc.
- g) **Source:** Source of the data should be stated.

Tabular presentation of data...

- **Frequency distribution**: is the organization of raw data in table form using classes and frequencies.
- **Frequency**: is the number of values in a specific class of the distribution
- **Raw data**: recorded information in its original collected form, whether it be counts or measurements, is referred to as raw data.

Frequency distribution (F.D.)...

- Frequency distribution can be **grouped or ungrouped**

The Ungrouped Frequency Distribution is also classified as:

- Ungrouped FD for Categorical Variables
- Ungrouped FD for Discrete Variables

1. Categorical Frequency Distribution

- Data are classified according to non-numerical categories.
- Categories must be mutually exclusive and exhaustive.
- Used to present nominal and ordinal data.

Ungrouped FD for Categorical Variables

A. Nominal data: Here the construction is straight forward: count the occurrences in each category and find the totals.

Example: The marital status of 60 adults classified as single, married, divorced and widowed is presented in a FD as below:

Marital status	Single	Married	Divorced	Widowed	Total
Frequency	25	20	8	7	60

Categorical F.D...

B. Ordinal data. The construction is identical to the nominal case. However, the categories should be put in an ordered manner.

Example: Satisfaction on teaching method in a class of size 80 is presented in a FD as shown below

Satisfaction	Very Satisfied	Satisfied	Dissatisfied	Very dissatisfied	Total
Frequency	15	36	3	7	60

Ungrouped FD for Discrete Variables

2. Ungrouped Discrete Frequency Distribution

- Count the number of times each possible value is repeated.

Example: In a survey of 30 families, the number of children per family was recorded and obtained the following data:

4 2 4 3 2 8 3 4 4 2 2 8 5 3 4 5 4 5 4 3 5 2 7 3
3 6 7 3 8 4.

Discrete/Ungrouped FD...

- These individual observations can be arranged in ascending order of magnitude to form an array: 2 2 2 2 2 3 3 3 3 3 3 3 4 4 4 4 4 4 4 4 5 5 5 5 6 7 7 8 8 8.
- The distribution of children in 30 families would be:

No.of children	2	3	4	5	6	7	8	Total
No. of family (f)	5	7	8	4	1	2	3	30

Continuous/grouped F.D

Continuous Frequency Distribution

- Arise from continuous variables/data.
- Unlike for a discrete FD, a class can not be allocated to each value of a continuous variable.
- Categories in to which the observations are distributed are called **classes** or **class intervals**.
- Classes should be exhaustive and mutually exclusive.

Continuous/grouped F.D...

Example: Consider the following FD on wages of 100 workers in a factory.

Wage (CI)	40-44	45-49	50-54	55-59	60-64	65-69	70-74	75-79
Freq.	6	9	15	17	20	13	12	8
CB's	39.5- 44.5	44.5- 49.5	49.5- 54.5	54.5- 59.5	59.5- 64.5	64.5- 69.5	69.5- 74.5	74.5- 79.5

□ workers earning between 40 and 44 birr (inclusive) are grouped in to the first class, workers earning between 45 and 49 birr (inclusive) are grouped in to the second class, and so on.

Continuous/grouped F.D...

Steps in constructing continuous frequency distribution

1. Determine the number of classes (k): Number of items belonging to a class.

- Decide k with the help of **Sturge's rule**:

$k = 1 + 3.322 \log n$, rounded up to the nearest integer.

Where $n \Rightarrow$ number of observations

$\log \Rightarrow$ common logarithm (logarithm of 10).

Example: If $n=10$, $k = 4.32 \approx 4$; if $n=100$, $k = 7.644 \approx 8$; if $n = 1000$, $k = 10.96 \approx 11$.

Continuous/grouped F.D...

2. Determine the Class Width (w): The difference between the successive class limits or class boundaries (may be upper or lower) of a class.

we use,
$$w \approx \frac{\text{Range}}{k}$$

Note that

“k”, rounded up to the nearest integer.

Continuous/grouped F.D...

3. Determine the Class Limits

- ▶ The lowest and height values that can be included in a class such that there is gap between successive classes.
- ▶ The lower class limit of the first class should be the smallest value of the observations.
- ▶ Add the size of a class on the lower class limit to obtain the lower class limit of the next higher classes.

Continuous/grouped F.D...

- ▶ To find the upper limit of the first class, subtract U from the lower limit of the second class.
- ▶ Then continue to add the class width to this upper limit to find the rest of the upper limits or
- ▶ Obtain the upper class limits by adding class width minus one to the corresponding lower class limits. i.e.
$$UCL = LCL + (W - 1)$$

Continuous/grouped F.D...

4. Determine the Class boundaries

- Let $U = \text{LCL of a class} - \text{UCL of preceding class}$. Add half of this difference ($U/2$) to all upper class limits to get the upper class boundaries (UCBs), and subtract ($U/2$) from all lower class limits to get the lower class boundaries (LCBs).

- $\text{UCBi} = \text{UCLi} + U/2$

- $\text{LCBi} = \text{LCLi} - U/2$

Where **Units of measurement (U)**: the distance between two possible consecutive measures. It is usually taken as 1, 0.1, 0.01, 0.001, -----.

Continuous/grouped F.D...

- 5. **Class mark (C.M) or Mid points:** it is the average of the lower and upper class limits or the average of upper and lower class boundary.
- 5. **Determine the frequency of each class:** determined simply by counting the number of observations belonging to each class.
- 6. **Cumulative frequency:** is the number of observations less than/more than or equal to a specific value.

Continuous/grouped F.D...

7. **Cumulative frequency above:** it is the total frequency of all values greater than or equal to the lower class boundary of a given class.
8. **Cumulative frequency below:** it is the total frequency of all values less than or equal to the upper class boundary of a given class.
9. **Relative frequency (rf):** it is the frequency divided by the total frequency.
10. **Relative cumulative frequency (rcf):** it is the cumulative frequency divided by the total frequency.

Example 1

The blood glucose level for 50 patients is shown below.

Construct a frequency distribution for the following data.

44	50	79	63	66	54	56	70	56	63
60	87	60	70	59	60	62	88	71	53
56	65	74	80	51	83	69	77	69	50
58	42	43	85	43	75	55	60	58	49
72	67	55	77	48	45	61	47	44	61

Solution:

Step 1: Find the highest and the lowest value $H=88$, $L=42$

Step 2: Find the range; $R=H-L=88-42=46$.

Step 3: Select the number of classes desired using Sturges formula;

$$k=1+3.322\log(50)=6.64=7(\text{rounding up})$$

Step 4: Find the class width; $w=R/k=46/7=6.57=7$ (rounding up)

Cont....

Step 5: Select the starting observation as lowest class limit (this is usually the lowest observation).

Add the width to that observation to get the lower limit of the next class. Keep adding until there are 7 classes.

42, 49, 56, 63, 70, 77, 84 are the lower class limits.

Step 6: Find the upper class limit; e.g. the first upper class=42-
 $U=49-1=48$

48, 55, 62, 69, 76, 83, 90 are the upper class limits.

So combining step 5 and step 6, one can construct the following classes.

So combining step 5 and step 6, one can construct the following classes.

Class limits

42-48

49-55

56-62

63-69

70-76

77-83

84-90

Step 7: Find the class boundaries by subtracting 0.5 from each lower class limit and adding 0.5 to the UCL as shown.

$$LCB_i = LCL_i - U/2$$

and $UCB_i = UCL_i + U/2$

Example: For class 1

$$LCB = 42 - 0.5 = 41.5 \text{ and}$$

$$UCB_1 = 48 + 0.5 = 48.5$$

Then continue adding W on both boundaries to obtain the rest boundaries. By doing so one can obtain the following classes.

Class boundary

41.5 – 48.5

48.5 – 55.5

55.5 – 62.5

62.5 – 69.5

69.5 – 76.5

76.5 – 83.5

83.5 – 90.5

Step 8: Tally the data.

Step 9: Write the numeric values for the tallies in the frequency column.

Step 10: Find cumulative frequency.

Step 11: Find relative frequency and /or relative cumulative frequency.

The complete frequency distribution follows

Class limits	Class boundary	Class Mark	Freq.	<CF	>CF	RF	<RCF	>RCF
42-48	41.5 – 48.5	45	8	8	50	0.16	0.16	1
49-55	48.5 – 55.5	52	8	16	42	0.16	0.32	0.84
56-62	55.5 – 62.5	59	13	29	34	0.26	0.58	0.68
63-69	62.5 – 69.5	66	7	36	21	0.14	0.72	0.42
70-76	69.5 – 76.5	73	6	42	14	0.12	0.84	0.28
77-83	76.5 – 83.5	80	5	47	8	0.10	0.94	0.16
84-90	83.5 – 90.5	87	3	50	3	0.06	1	0.06
Total			50			1		

Types of table

- Simple or one-way table:
- Two-way table:
- Higher Order Table
- Super tables

Simple or one-way table

- It is also called simple frequency table.
- when the individual observations involve only to a single variable
- A single variable is presented with respective frequency (counts or percentages) of each of the categories.

e.g. Table 2: Socio-demographic characteristics of respondents, DBCSH, North Shoa, Ethiopia, 2022.

Variable	Category	Frequency (n=1514)	Relative frequency (%)
Sex	Male	986	65.1
	Female	528	34.9

Two-way table

- Presents two variables cross-tabulated in a single table.
- It is important to examine the relationship between two variables.
- Also known as a contingency table.
- Epidemiologists use it in calculating measures of association and tests of statistical significance.
- These tables are convenient for comparing persons with and without the exposure, and those with and without the disease.

		Disease	
Exposure		Yes (+)	No (-)
Yes (+)	a	b	a + b
No (-)	c	d	c + d

Higher Order Table

- You may want to include a third variable to show a set of data more completely.
- e.g. three-variable table for cigarette smoking, sex and lung cancer

History of cigarette smoking/sex		Lung cancer		
		Yes	No	Total
Yes	Male			
	Female			
No	Male			
	Female			
Total				

Table 2: Sociodemographic characteristics of reproductive-age women in Kenya using KDHS 2022

Super Tables

- A great deal of information can be presented in a closely related group of tables called a super table.
- *Table 1: Sociodemographic characteristics of reproductive-age women in Kenya using KDHS 2022*

Variables		Weighted frequency (%)
Age	15-19	414 (1.12%)
	20-24	2621 (7.03%)
	25-29	5844 (15.64%)
	30-34	7030 (18.86%)
	35-39	8447(22.65%)
	40-44	6685(17.94%)
	45-49	6244(16.76%)
Religion	Catholic	6298(16.89%)
	Protestant	13,522(36.27%)
	evangelical churches	9376(25.15%)
	African instituted churches	3335(8.95%)
	Islam	3126(8.39%)
09/25/2026	Others	1628(4.37%)
	Marital status	
	Not married	11,076(29.71%)

2. Diagrammatic Representation

What is the difference between diagram and figure?

Importance of diagrammatic representation:

- ✚ To understand the information easily.
- ✚ To make the data attractive.
- ✚ To make comparisons of items easy.
- ✚ To draw attention of the observer.
- ✚ The purpose of graphs and diagrams is not to provide exact and detailed information, but simple comparisons.
- ✚ Any further information shall rather be obtained from the original data
- ✚ When graphs are poorly designed, they not only ineffectively convey message, but they are often misleading.

Graphical Presentation...

Limitations of Graphical presentation

- The technique of diagrammatic presentation is made use only for **purposes of comparison**.
- It is not to be used when comparison is either not possible or is not necessary.
- Diagrammatic presentation is **not an alternative to tabulation**.
- It **only strengthens** the textual exposition of a subject, and cannot serve as a complete substitute for statistical data.

Graphical Presentation...

- It can give only an **approximate idea** and as such where greater accuracy is needed diagrams will not be suitable.
- They fail to bring to light small differences

GENERAL DIRECTIONS FOR THE CONSTRUCTION OF DIAGRAMS

- ◆ The first thing is the selection of a **proper scale**.
- ◆ All the significant characteristics of the figures should be clearly exhibited by the diagram, and it should also suit the size of the paper.
- ◆ The **vertical** and **horizontal** scales should be clearly shown on the diagram itself - the former on the **left** hand side and the latter at the **bottom** of the diagram.
- ◆ **Neatness** should be strictly observed and the diagram be drawn with the aid of geometrical instruments

Graphical presentation cont...

- ♣ The heading (title) should be written in **bold letter** and should be **self explanatory**.

The source should be indicated if the data are not collected by yourself.

- ♣ Various shades of colours can be used.
- ♣ Diagrams should be **as simple as possible** so that the reader can understand their meaning clearly and easily.

N.B. The graph must present a **truthful impression of the data**

Specific types of graphs include:

Bar graph

Pie chart



**Nominal, ordinal
data**

Histogram

Stem-and-leaf plot

Box plot

Scatter plot

Line graph

Others



**Quantitative
data**

1. BAR CHARTS (OR GRAPHS)

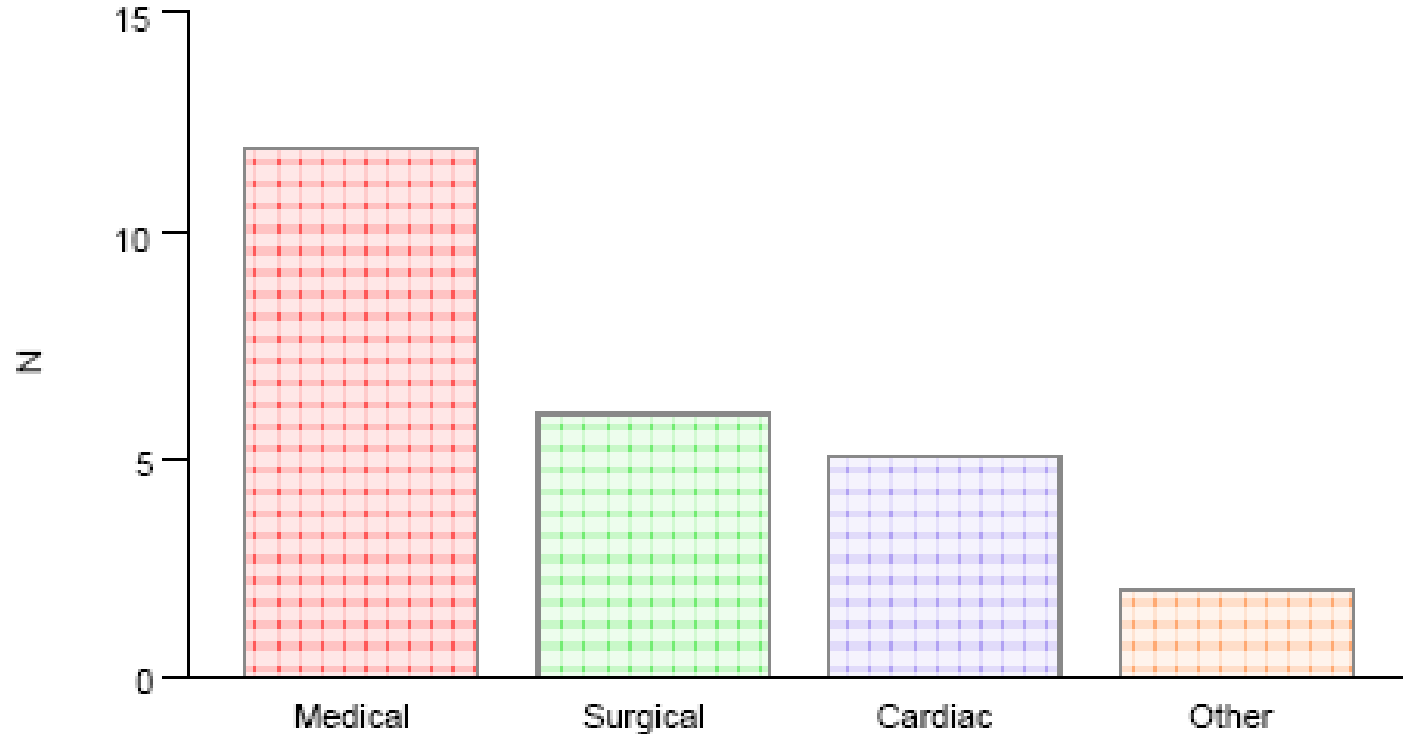
- Bar graph is especially satisfactory for nominal and ordinal data.
- Categories are listed on the horizontal axis (X-axis)
- Frequencies or relative frequencies are represented on the Y-axis (ordinate).
- The heights of bars represent the value of the frequency (actual number or percentage) for each category.

Tips for constructing Bar graph

- ⊕ Whenever possible it is better to construct a bar diagram on a graph paper
- ⊕ All bars drawn in any single study should be of the same width
- ⊕ The different bars should be separated by equal distances
- ⊕ All the bars should rest on the same line called the base
- ⊕ Whenever possible, it is advisable to draw bars in order of magnitude

2.1. Simple bar chart: It is a one-dimensional diagram

E.g. **Bar chart for the type of ICU for 25 patients**

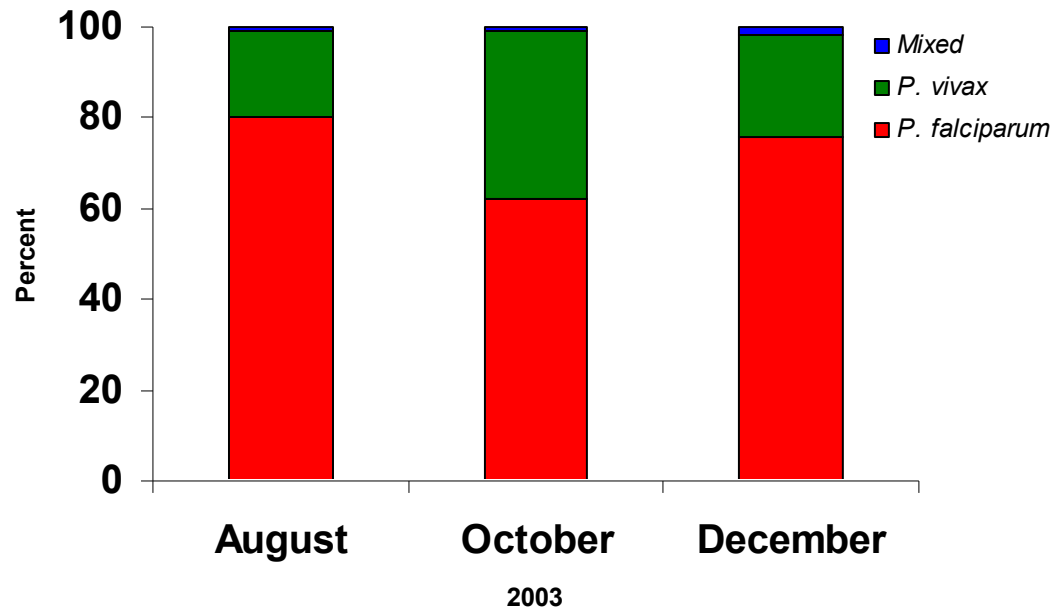


2.2 Sub-divided bar chart

- If there are different quantities forming the sub-divisions of the totals, simple bars may be sub-divided in the ratio of the various sub-divisions to exhibit the relationship of the parts to the whole.
- The order in which the components are shown in a “bar” is followed in all bars used in the diagram.

–**Example:** Stacked and 100% Component bar charts

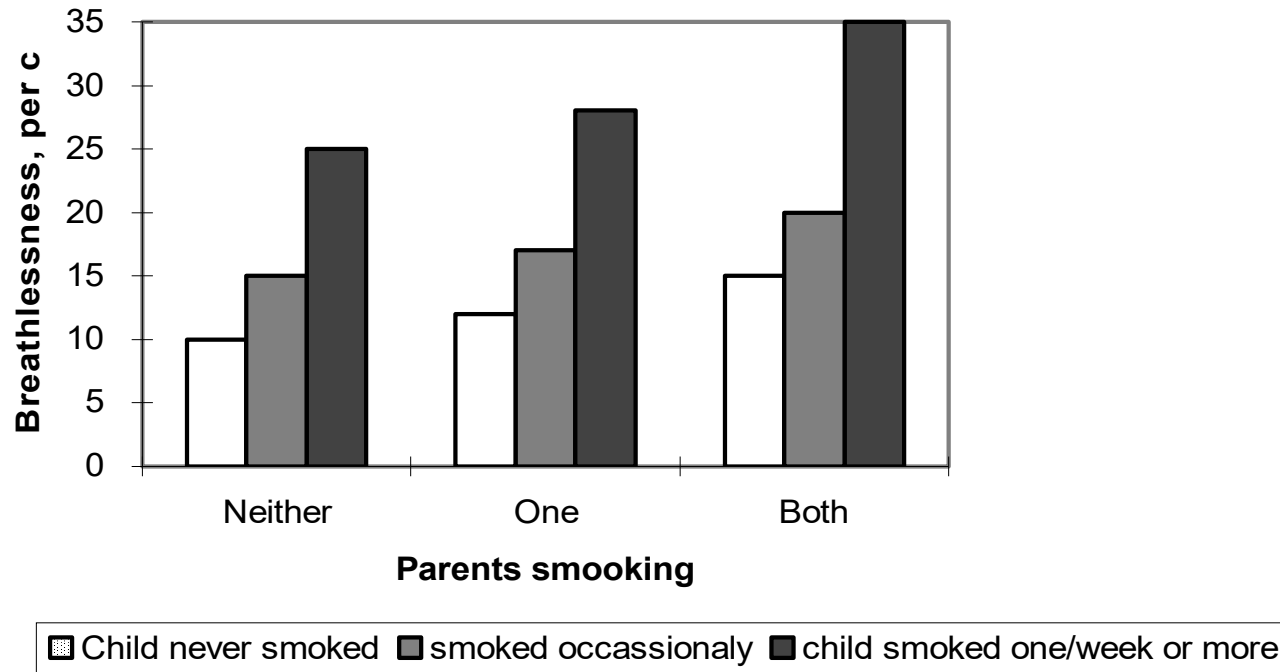
Example: Plasmodium species distribution for confirmed malaria cases, Zeway, 2003



2.3. Multiple bar graph

- Bar charts can be used to represent the *relationships* among more than two variables.
- The following figure shows the relationship between children's reports of breathlessness and cigarette smoking by themselves and their parents.

Prevalence of self reported breathlessness among school children, 1998



We can see from the graph quickly that the prevalence of the symptoms increases both with the child's smoking and with that of their parents.

2. Pie chart

- Shows the **relative frequency** for each category by dividing a circle into sectors, the angles of which are proportional to the relative frequency.
- Used for a **single categorical** variable
- Pie chart is important for depicting **discrete variables with relatively few categories**.

Relative Frequency	Size of Wedge, in degrees
0.50	50% of 360 = 180 degrees
0.25	25% of 360 = 90 degrees
p	$(p) \times (100\%) \times 360 \text{ degrees}$

Steps in constructing pie chart

- ▶ Steps in constructing a pie-chart
 - ▶ 1. Construct a frequency table
 - ▶ 2. Change the frequency into percentage (P) or fraction (F)
 - ▶ 3. Change the percentages into degrees, where:
 - ▶ **Angle of sector** = $\frac{\text{component part}}{\text{total}} * 360$
 - ▶ 4. Draw a circle and divide it accordingly

pie chart for severity_level_of depression in %

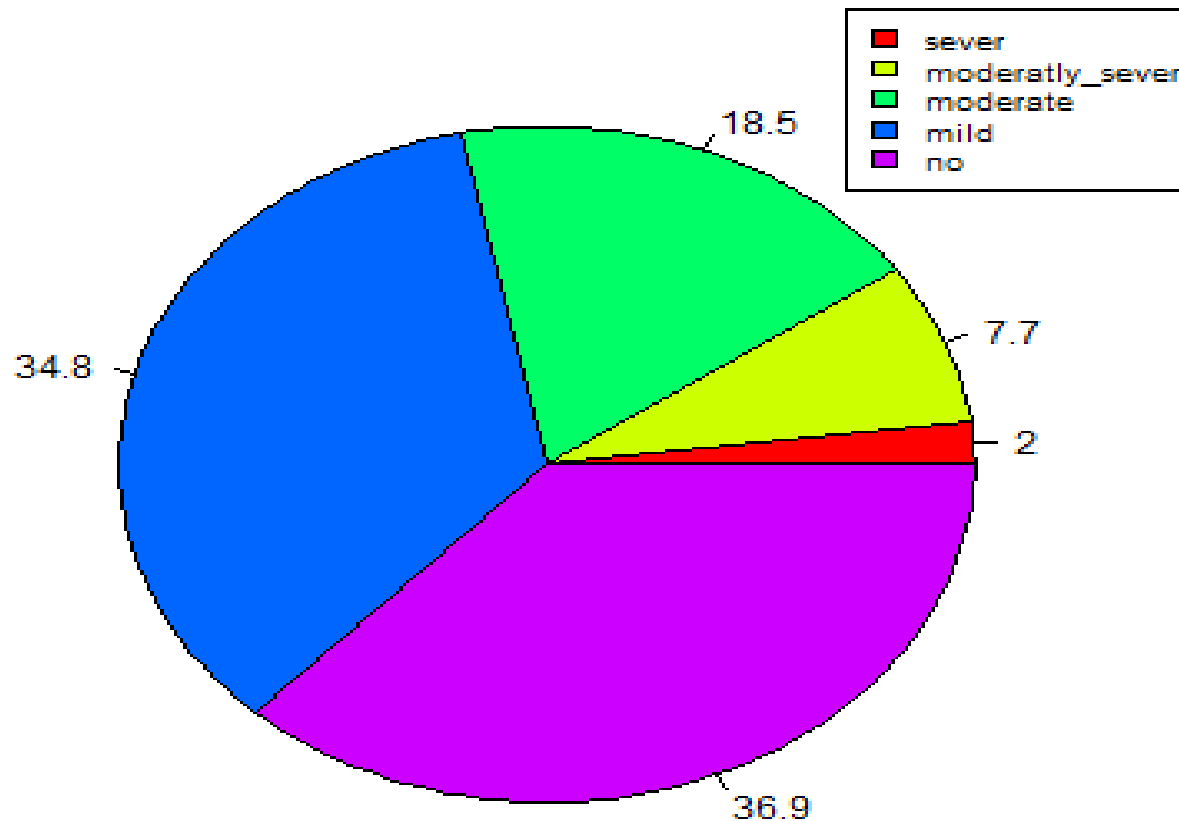


Figure 1: proportion of depression severity level among high school and preparatory school adolescents in Gondar town 2022.

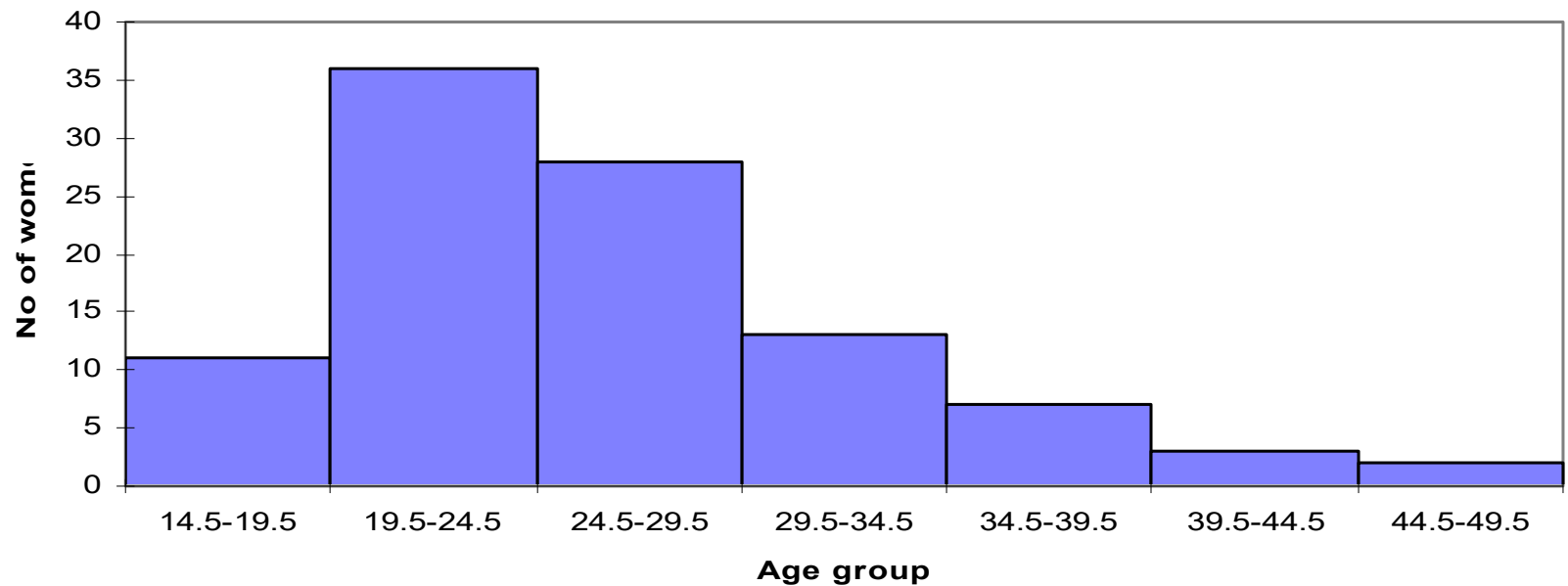
3. Histogram

- Histograms are frequency distributions with **continuous class intervals** that have been turned into graphs.
- To construct a histogram, we draw the **class boundaries** on a horizontal line and the frequencies on a vertical line.
- Non-overlapping intervals that cover all of the data values must be used.
- Bars are drawn over the intervals in such a way that the areas of the bars are all proportional in the same way to their interval frequencies.

Example: Distribution of the age of women at the time of marriage

Age Group	15-19	20-24	25-29	30-34	35-39	40-44	45-49
Number	11	36	28	13	7	3	2

Age of women at the time of marriage



Histogram for weight of infants

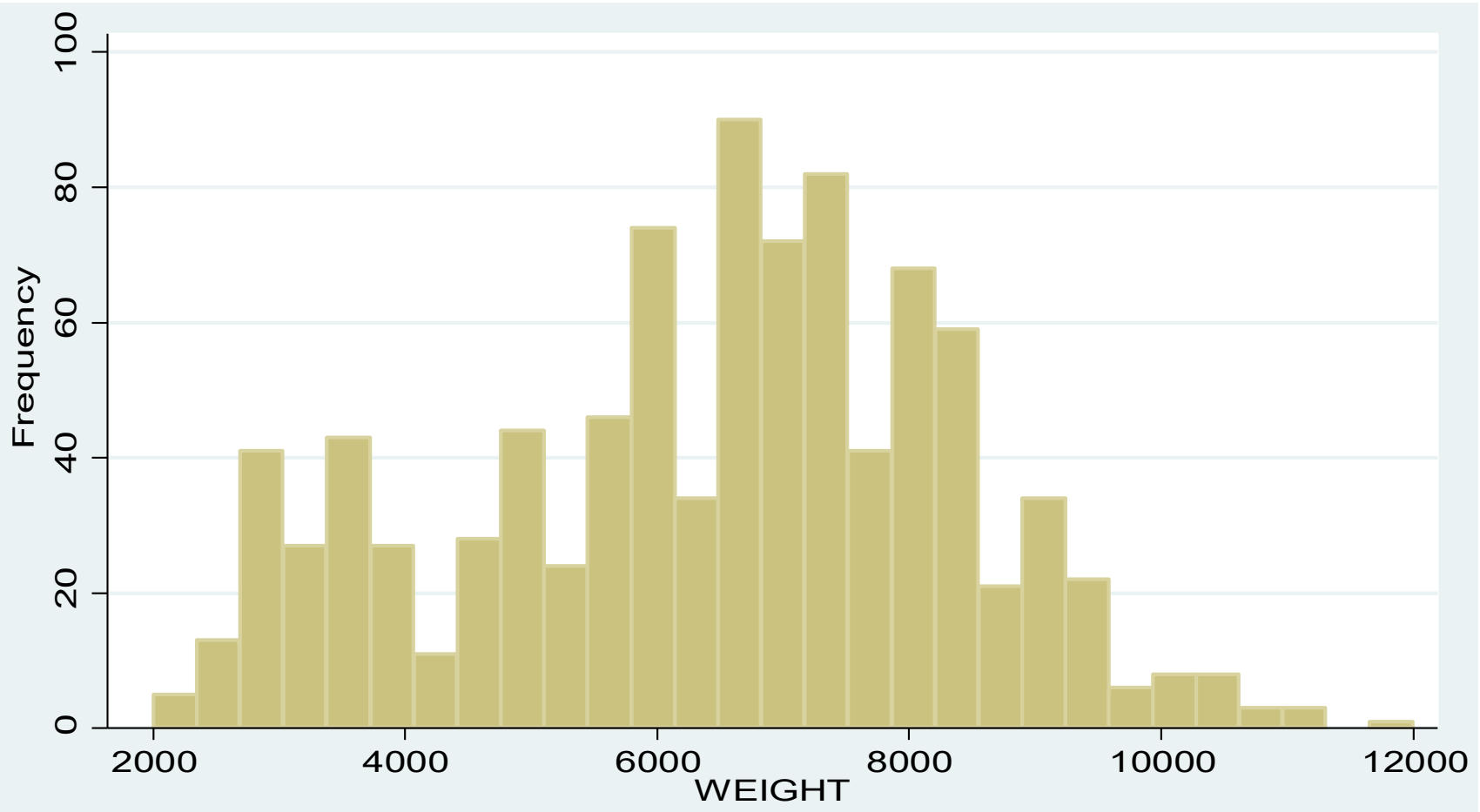


Figure 1 :- shows histogram of weight of 150 infants during their first year, Ethiopia, 2015

Histogram...

Two problems with histograms

1. They are somewhat difficult to construct
2. The actual values within the respective groups are lost and difficult to reconstruct

⇒ ***stem-and-leaf plot overcomes these problems***

4. Stem-and-Leaf Plot

- A quick way to organize data to give visual impression similar to a histogram while retaining much more detail on the data.
- Similar to histogram and serves the same purpose and reveals the presence or absence of symmetry
- Are most effective with relatively small data sets
- Are not suitable for reports and other communications, but
- Help researchers to understand the nature of their data

Example

- 43, 28, 34, 61, 77, 82, 22, 47, 49, 51, 29, 36, 66, 72, 41

2		2	8	9
3		4	6	
4		1	3	7 9
5		1		
6		1	6	
7		2	7	
8		2		

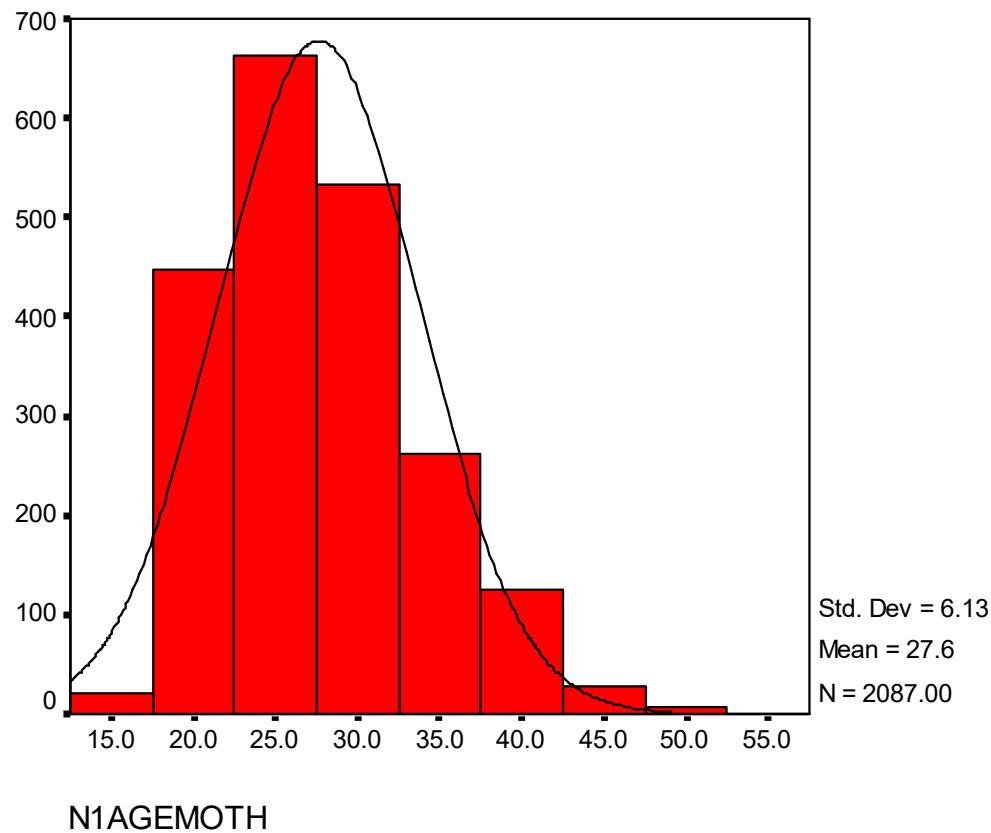
Example: 3031, 3101, 3265, 3260, 3245, 3200, 3248,
3323, 3314, 3484, 3541, 3649 (BWT in g)

Stem	Leaf	Number
30	<u>31</u>	1
31	<u>01</u>	1
32	<u>65</u> <u>60</u> <u>45</u> <u>00</u> <u>48</u>	5
33	<u>23</u> <u>14</u>	2
34	<u>84</u>	1
35	<u>41</u>	1
36	<u>49</u>	1

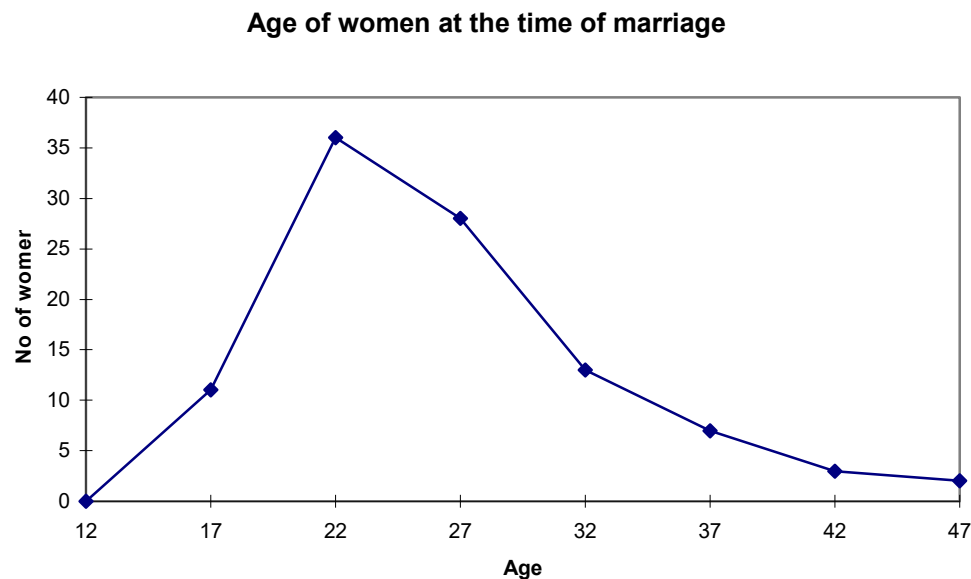
5. Frequency polygon

- A frequency distribution can be portrayed graphically in yet another way by means of a frequency polygon.
- To draw a frequency polygon we connect the mid-point of the tops of the cells of the histogram by a straight line (i.e. By connecting mid point of the class boundary)
- The total area under the frequency polygon is equal to the area under the histogram
- Useful when comparing two or more frequency distributions by drawing them on the same diagram

Frequency polygon for the ages of 2087 mothers with <5 children, Adami Tulu, 2003



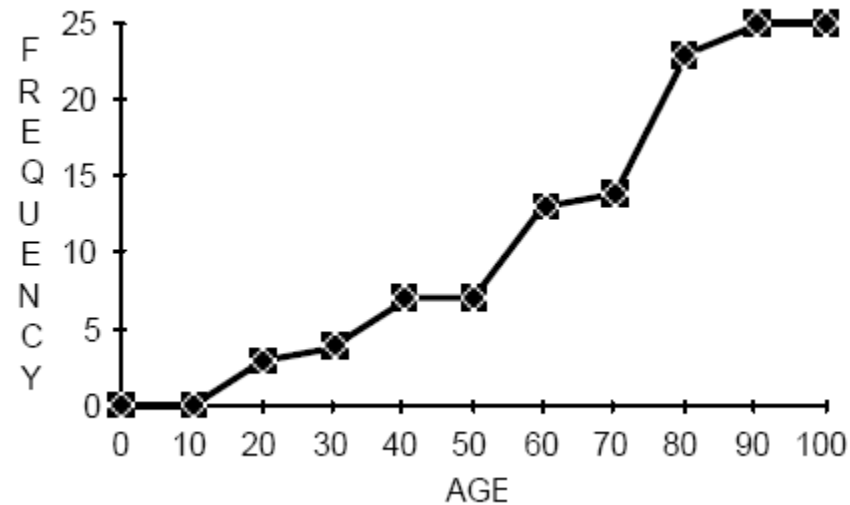
- It can be also drawn without erecting rectangles by joining the top midpoints of the intervals representing the frequency of the classes as follows:



6. Ogive Curve (The Cumulative Frequency Polygon)

- Some times it may be necessary to know the number of items whose values are **more or less than a certain amount**.
- We may, for example, be interested to know the no. of patients whose weight is <50 Kg or >60 Kg.
- To get this information it is necessary to change the form of the frequency distribution from a 'simple' to a 'cumulative' distribution.
- **Ogive curve** turns a cumulative frequency distribution in to graphs.

Cumulative frequency of 25 ICU patients



7. Scatter plot...

- Most studies in medicine involve *measuring more than one characteristic*, and graphs displaying the relationship between two characteristics are common in literature.
- When both the variables are qualitative then we can use a *multiple bar graph*.
- When one of the characteristics is qualitative and the other is quantitative, the data can be displayed in *box and whisker plots*.

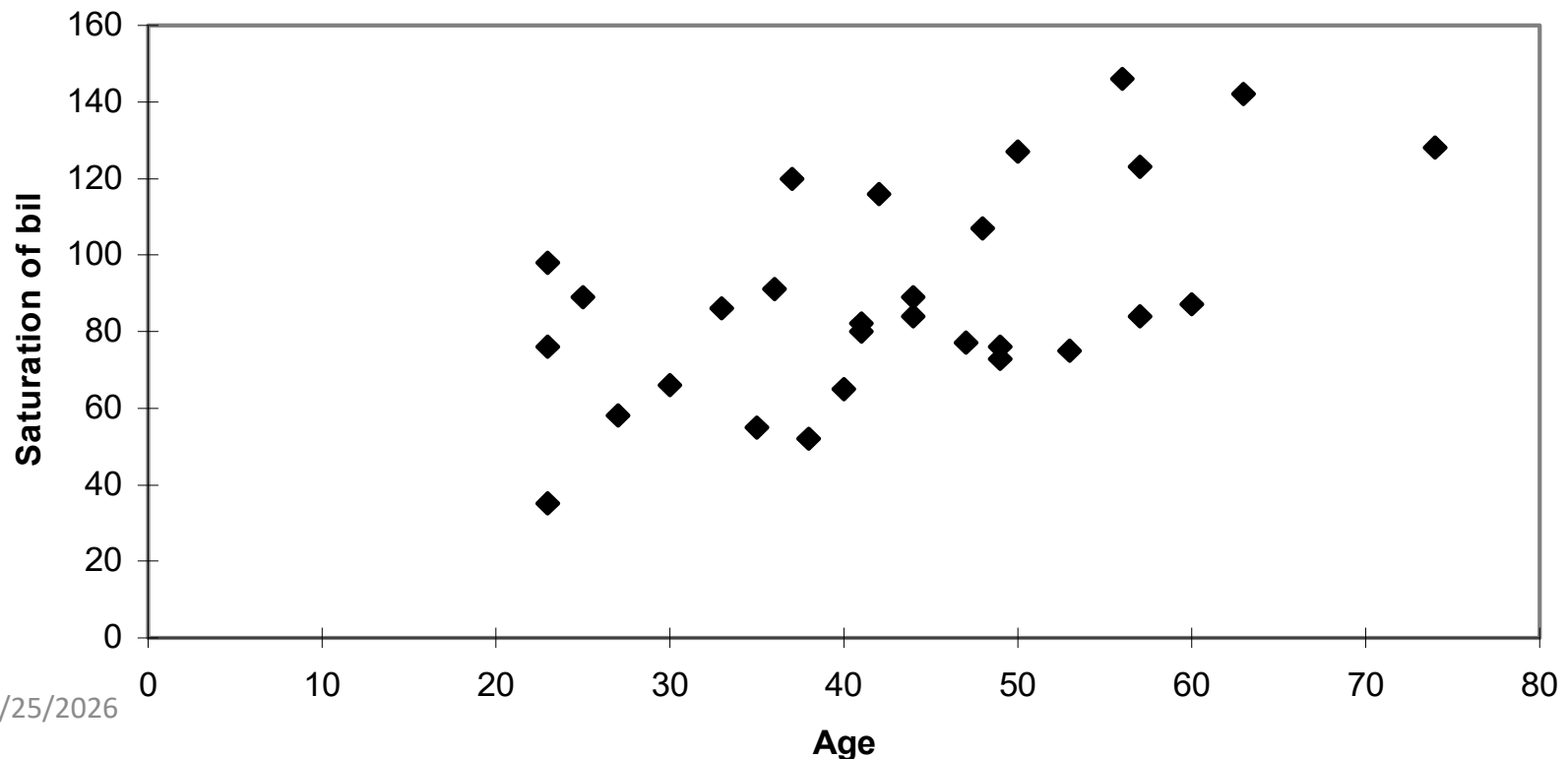
Scatter plot...

For two quantitative variables we use bivariate plots (also called scatter plots or scatter diagrams).

In the study on percentage saturation of bile, information was collected on the age of each patient to see whether a relationship existed between the two measures.

- A scatter diagram is constructed by drawing X-and Y-axes.
- Each point represented by a point or dot(•) represents a pair of values measured for a single study subject

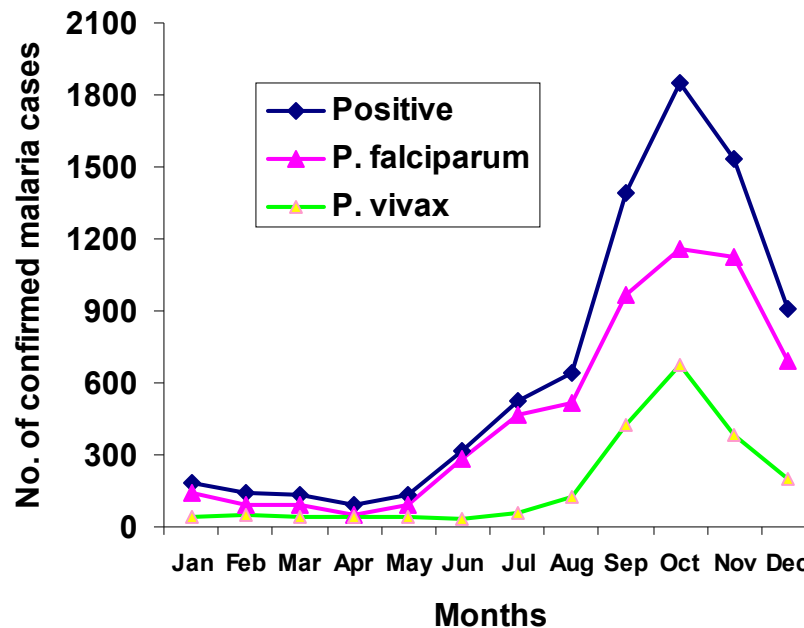
Age and percentage saturation of bile for women patients in hospital Z, 1998



8. Line graph

- Useful for assessing the **trend** of particular situation overtime.
- Helps for monitoring the trend of epidemics.
- The time, in weeks, months or years, is marked along the horizontal axis, and
- Values of the quantity being studied is marked on the vertical axis.
- Values for each category are connected by continuous line.

No. of microscopically confirmed malaria cases by species and month at Zeway malaria control unit, 2003



Reading assignment

- **Pareto chart ??**
- **Q-Q plot ??**
- **Dependent and independent variables for Qualitative research?**
- **Time as interval and as a ratio scale ?**

THANK YOU!!!